



NATIONAL UNIVERSITY OF SCIENCES & TECHNOLOGY

Department of Electrical Engineering

INTERNSHIP REPORT

National Electric Power Regulatory Authority (NEPRA), Islamabad

Submitted by

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3rd Year, BE Electrical Engineering | NUST

Internship Period

06 July 2026 - 14 August 2026 (6 Weeks)

Departments: Information Technology (IT) | Technical

Acknowledgement

I would like to express my sincere gratitude to the National Electric Power Regulatory Authority (NEPRA), Islamabad, for providing me the opportunity to complete my internship with the organization. I am especially grateful to Mr. Attar Rehman, Deputy Director (Technical), and Mr. Muhammad Akram, Deputy Director (IT), for supervising my work in their respective departments and for their guidance throughout the internship.

The exposure to both the Information Technology and Technical departments gave me a practical understanding of how a federal regulatory body operates and how engineering knowledge is applied in a real regulatory and utility context.

I am also thankful to my faculty supervisor at NUST for the academic support in documenting this experience.

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1. Introduction

1.1 About NEPRA

The National Electric Power Regulatory Authority (NEPRA) is the federal regulatory body responsible for overseeing Pakistan's electric power sector. Established under the NEPRA Act, 1997, the Authority regulates generation, transmission, and distribution licensees; determines tariffs; monitors compliance and performance standards across the power sector; and issues an annual State of the Industry Report that reviews the sector's performance and highlights key regulatory issues.

1.2 Objectives of the Internship

- Gain first-hand exposure to the operations of a federal energy sector regulator.
- Develop practical, hands-on experience in web technologies and website management (IT Department).
- Build technical understanding of Advanced Metering Infrastructure (AMI) and power transformers (Technical Department).
- Study and analyze official regulatory publications, in particular the State of the Industry Report 2025.
- Apply academic knowledge of electrical engineering in a real-world regulatory and utility setting.

1.3 Internship Duration & Schedule

The internship was carried out over 6 weeks, from 06 July 2026 to 14 August 2026, with working hours from 9:30 AM to 3:30 PM. The first two weeks were spent in the IT Department, followed by four weeks in the Technical Department, as summarized below.



Figure 1.1 - Internship schedule by department

2. IT Department (Weeks 1-2)

The first two weeks of the internship were spent in the IT Department, where the focus was on the NEPRA official website: its content, search visibility, and functional reliability.

2.1 NEPRA Official Website

I was introduced to the structure and content management workflow of the NEPRA official website. This included understanding how pages are organized, how content is published, and how the site is maintained on an ongoing basis.

2.2 Google Search Console & Website Indexing

A significant part of this phase involved learning to use Google Search Console, which is Google's tool for webmasters to submit a website, check whether individual pages are indexed, and request indexing of new or updated pages.

Indexing is the process by which Google adds a webpage to its search index, which is what allows that page to subsequently appear in search results. Without indexing, a page can exist on a live website but remain effectively invisible to anyone searching for it on Google. Using Search Console, I checked the indexing status of NEPRA website pages, submitted sitemaps, and requested indexing for pages that had not yet been picked up by Google's crawlers.

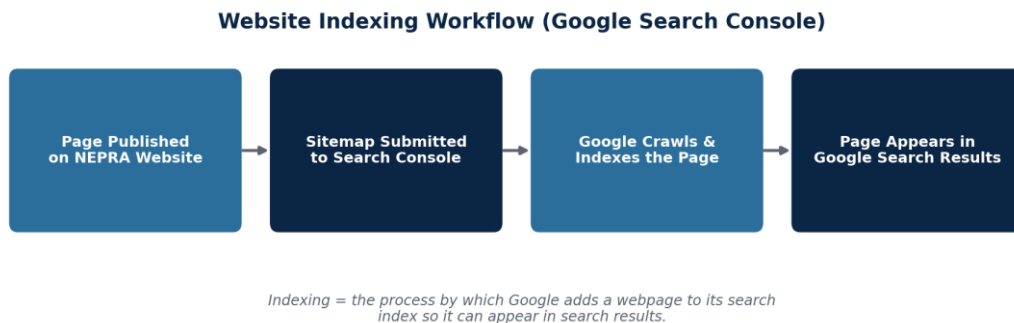


Figure 2.1 - Website indexing workflow via Google Search Console

2.3 Website Debugging

Alongside indexing work, I was involved in identifying and debugging issues on the live NEPRA website, such as broken links, page rendering errors, and other functional bugs, to improve the overall reliability and usability of the site for the public and for regulated entities that use it to access licensing and tariff information.

2.4 NEPRA Document Intelligence Portal

As a further IT Department task, I worked on a document intelligence portal for NEPRA - a web-based tool for centralizing and searching the Authority's regulatory filings (tariffs, licenses, determinations, and reports). The portal supports three main ways of getting documents in: direct upload (PDF, DOCX, XLSX, PPTX, TXT, CSV, MD - up to 20 files at once), a web crawler that pulls document links directly from a NEPRA or regulator web page, and indexed full-text search across the archive.

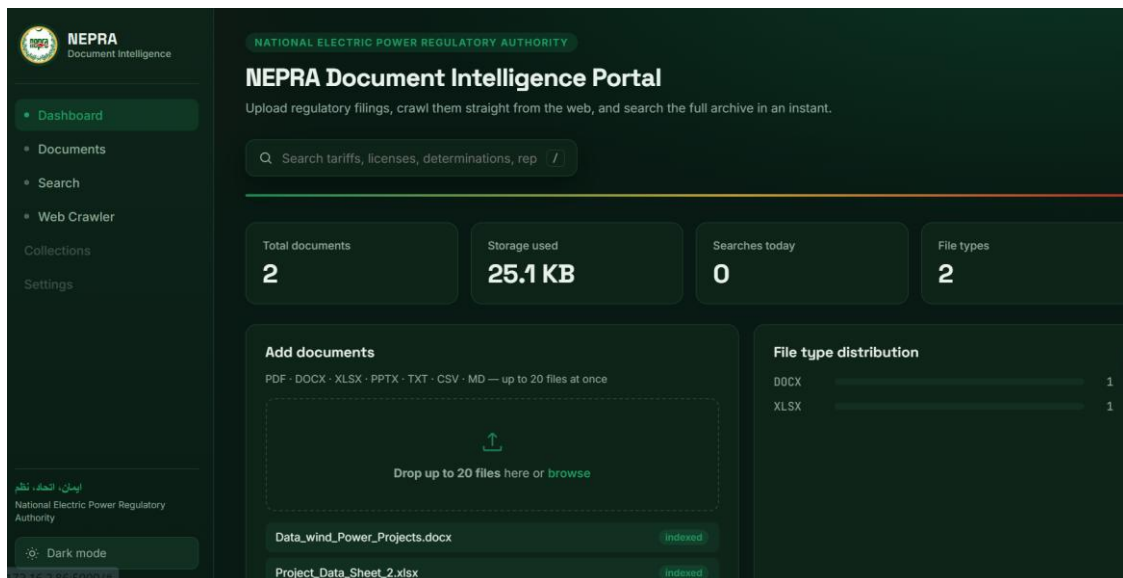


Figure 2.2 - NEPRA Document Intelligence Portal - dashboard, upload, and crawler view

Once uploaded or crawled, each document is indexed and listed with key metadata - file type, size, page count, word count, and auto-generated tags - making it possible to quickly locate and reference specific filings instead of searching through folders manually.

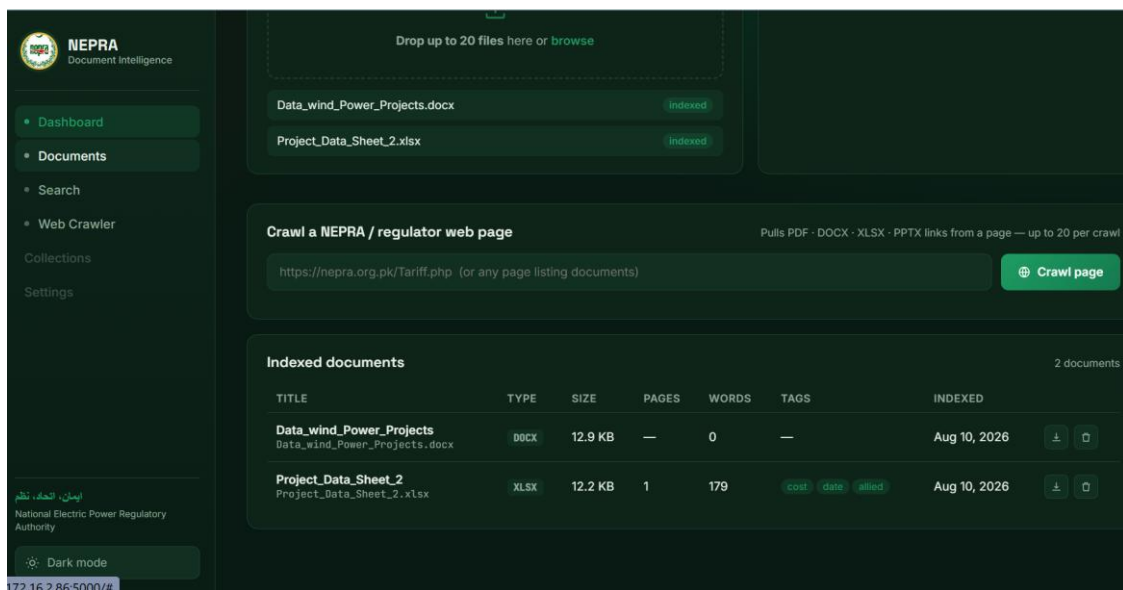


Figure 2.3 - Indexed documents view, showing file metadata and tags

3. Technical Department (Weeks 3-6)

The remaining four weeks were spent in the Technical Department, where the focus shifted to core power system components and to regulatory analysis of the sector, using NEPRA's State of the Industry Report 2025 as the primary reference document.

3.1 Advanced Metering Infrastructure (AMI)

I studied Advanced Metering Infrastructure (AMI) meters: their components, communication architecture, and role within a modern distribution network. AMI meters differ from conventional meters in that they support two-way communication, enabling remote reading, load profiling, and outage detection, rather than requiring manual, on-site meter reading.

AMI System - Functional Block Diagram

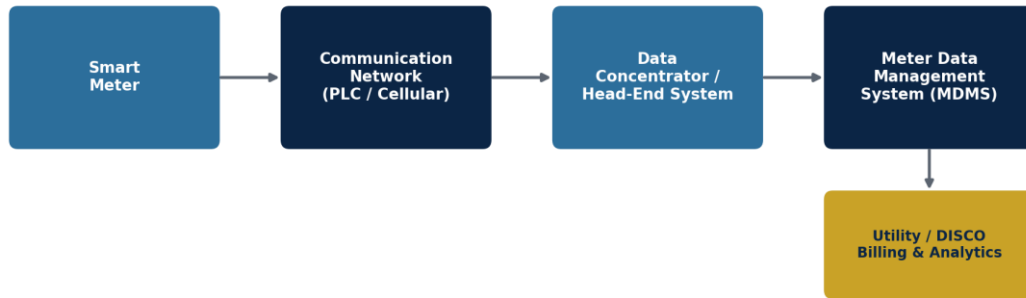


Figure 3.1 - AMI system functional block diagram

The table below summarizes the main components of an AMI meter and their functions, as covered during this phase of the internship:

Component	Function
Metering Unit	Measures voltage, current, energy consumption and power quality parameters
Communication Module	Transmits meter data via PLC (Power Line Carrier) or cellular network
Memory / Data Logger	Stores interval consumption data (load profile) locally on the meter
Display Unit	Shows real-time readings for on-site verification
Tamper Detection	Flags unauthorized access or meter interference
Battery Backup	Maintains clock and logging function during power outages

This study was directly relevant to ongoing regulatory activity: NEPRA's State of the Industry Report 2025 notes that during FY 2024-25, all DISCOs approached the Authority for approval to invest in AMI meter rollout across Pakistan. The report also flags that in areas where AMI/AMR meters have already been installed in limited numbers, integration has often been incomplete, with recurring cases of non-communicating ("mute") meters, and it recommends that a clear, publicly shared rollout roadmap and DISCO capacity-building accompany any large-scale deployment.

3.2 Transformers

I studied the construction and working principle of power transformers, covering the core, windings, insulation system, and cooling arrangement, as well as transformer ratings, types, and typical loss mechanisms (copper losses and core/iron losses).

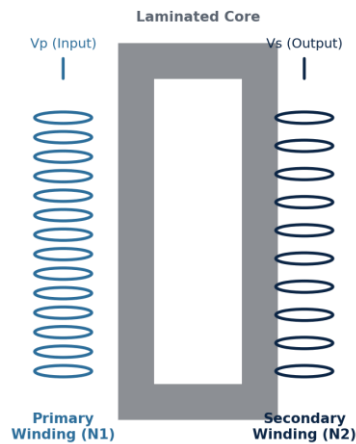
Single-Phase Core-Type Transformer - Basic Construction

Figure 3.2 - Basic construction of a single-phase core-type transformer

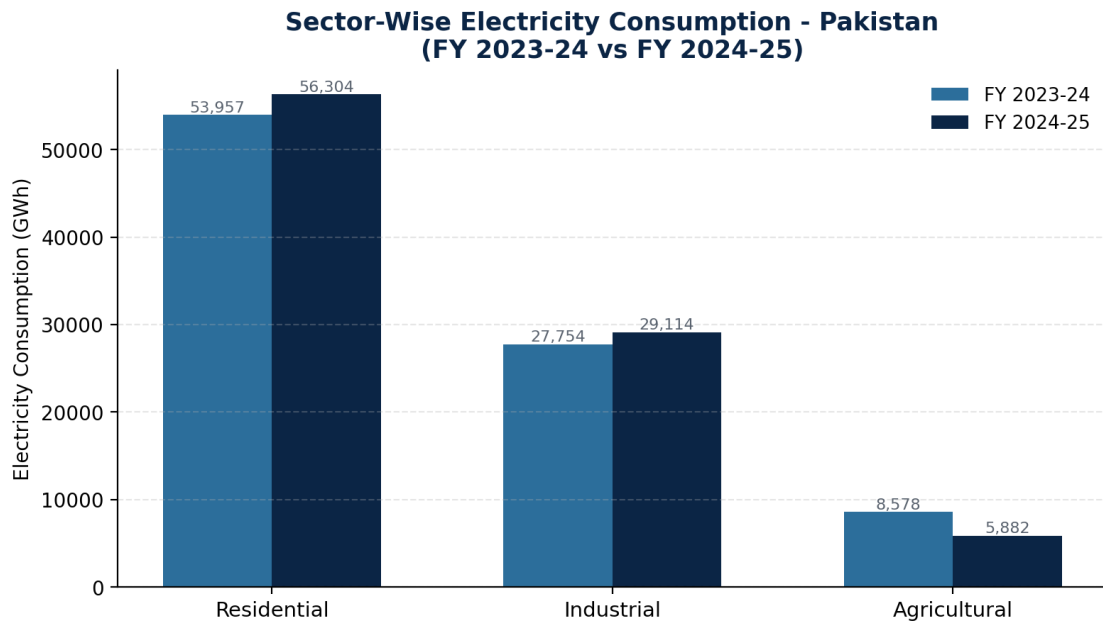
Component	Function
Core (Laminated Steel)	Provides a low-reluctance path for magnetic flux; laminated to reduce eddy current losses
Primary Winding	Receives input (supply-side) voltage and creates the magnetic flux
Secondary Winding	Induced with the output voltage as per the turns ratio ($N1/N2 = V1/V2$)
Insulation	Separates windings/core electrically and withstands operating voltage stress
Tank & Cooling System (Oil/Air)	Houses the core-coil assembly and dissipates heat generated during operation
Bushings	Insulated terminals that bring winding leads out of the tank safely

Later in this phase, the discussion extended to transformer protection and testing practices used to verify transformer condition and safe operation before and during service.

3.3 State of the Industry Report 2025 - Chapter 2 Review

The final part of the Technical Department phase involved a detailed review of Chapter 2 ("Overview of the Electric Power Sector") of NEPRA's State of the Industry Report 2025, which covers the generation, transmission, and distribution segments of Pakistan's power sector for FY 2024-25. Key themes reviewed included generation cost dynamics, transmission project delays, DISCO privatization, and the decline in overall electricity demand.

One notable data point examined was the change in national electricity consumption by sector between FY 2023-24 and FY 2024-25, summarized below:



Source: NEPRA State of the Industry Report 2025, Section 2.3.8

Figure 3.3 - Sector-wise electricity consumption, FY 2023-24 vs FY 2024-25 (Source: State of the Industry Report 2025, Sec. 2.3.8)

Total electricity consumption across all DISCOs and KE stood at 111,466.69 GWh in FY 2024-25, a 1.71% increase over 109,591.26 GWh in FY 2023-24, though still below the FY 2021-22 level. The residential sector's share of total consumption rose from about 49.23% to 50.51%, while agricultural consumption fell sharply, down roughly 31.4% year-on-year. The report attributes this pattern less to genuine growth in household demand and more to a steeper decline in agricultural and, to a lesser extent, industrial usage, driven by high electricity tariffs pushing consumers toward conservation and solar alternatives.

This review reinforced the connection between the technical concepts studied earlier in the internship (AMI, transformers) and the regulatory/financial issues NEPRA monitors at a sector-wide level, such as distribution losses, circular debt, and the pace of metering modernization.

3.4 NTDC / National Grid Company (NGC) - Transmission Network

As an extension of the Chapter 2 review, I also studied Chapter 4 of the State of the Industry Report 2025, which covers the performance of Pakistan's transmission network. The former National Transmission and Despatch Company (NTDC) now operates under the name National Grid Company of Pakistan Limited (NGC), licensed under Section 17 of the NEPRA Act to transmit power at 220 kV and above across the country, excluding Karachi (which is served by KE's own network).

As of 30 June 2025, NGC's transmission infrastructure comprised the following:

Voltage Level	Grid Stations	Transformation Capacity	Circuits	Circuit Length
500 kV	19	25,950 MVA	86	9,100 km
220 kV	49	30,850 MVA	158	11,853 km

A combined 233 power/auto-transformers are installed across NGC's 500 kV and 220 kV grid stations. The report also notes that, in FY 2024-25, 116 of the 192 transformers spread across 68 grid stations were operating above 80% of rated capacity, with recurring overloading reported at grid stations such as Sheikhpura, Gatti, Multan, and Jamshoro, contributing to reliance on costlier generation and constrained power evacuation from the south.

NGC's Transmission and Transformation (T&T) losses are tracked monthly against a NEPRA-approved threshold of 2.50%, set as part of the FY 2022-23 to FY 2024-25 investment plan. Actual losses exceeded this threshold for much of the second half of FY 2024-25, as shown below:

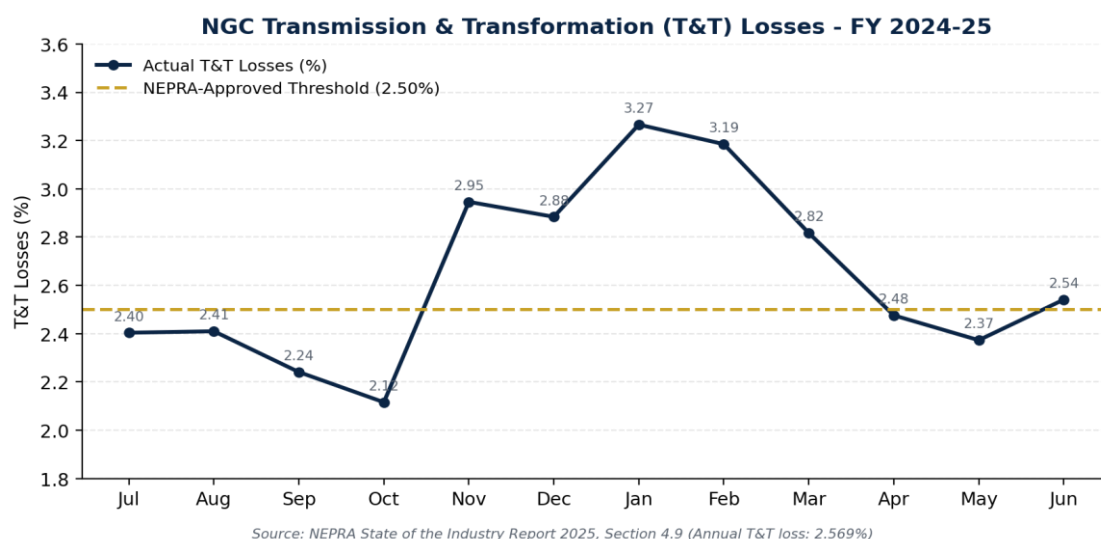


Figure 3.4 - NGC monthly T&T losses vs NEPRA-approved threshold, FY 2024-25 (Source: State of the Industry Report 2025, Sec. 4.9)

Annual T&T losses for FY 2024-25 came out to 2.569% (3,432.19 GWh lost out of 133,621.29 GWh received), against an approved level of 2.50% and an actual FY 2023-24 figure of 2.524%. Under its NEPRA-approved investment plan, NGC's allowed capital expenditure across the three-year period totaled roughly Rs. 352.3 billion, split as follows:

Fiscal Year	Grand Total Allowed (Rs. Million)	T&T Loss Allowed	T&T Loss Actual
FY 2022-23	118,515	2.422%	2.422%
FY 2023-24	113,124	2.500%	2.524%
FY 2024-25	120,683	2.500%	2.569%
Total	352,322	-	-

Studying this section highlighted how transmission-level performance (losses, overloading, and delayed projects such as the 500 kV Lahore-North Grid Station and SCADA-III modernization) feeds directly into the cost and reliability issues discussed at the generation and distribution level in Chapter 2, giving a more complete, end-to-end picture of the power sector NEPRA regulates.

3.5 NEPRA Licensing & Tariff-Setting Framework

In the final week, my technical-department supervisor walked me through how NEPRA's two core regulatory functions - licensing and tariff-setting - fit together, using the Licensing and Regulations sections of NEPRA's official Legal page as reference. The overall flow is straightforward: a company must first obtain a license for the electric power service it wants to provide, a tariff is then determined for that license, and only then can the licensee sell electric power.

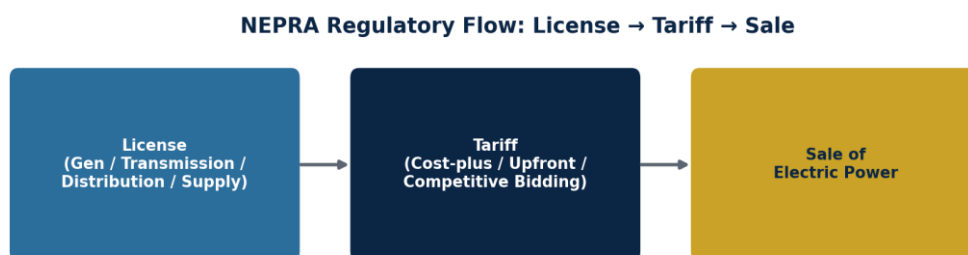


Figure 3.5 - NEPRA's regulatory flow from licensing through to tariff and sale

NEPRA issues separate licenses for each electric power service - Generation, Transmission (including the National Grid Company and Special Purpose Transmission), Distribution, and Electric Power Supply. Under more recent reforms, the Supply License category itself has been split into Competitive Suppliers, On-site/Captive Suppliers, and a Supplier of Last Resort (SOLR), reflecting NEPRA's move toward a more competitive retail supply market. The licensing process itself - application, eligibility review, advertisement for public comment, and the Authority's decision - is set out in the NEPRA Licensing (Application, Modification, Extension and Cancellation) Procedure Regulations, 2021.

Generation tariffs in particular can be set through three different mechanisms, summarized below:

Mechanism	How it works
Cost-Plus Tariff	Tariff is built up from the project's actual capital and operating costs, reviewed through a "prudence check" by NEPRA on a case-by-case basis before approval.
Upfront Tariff	NEPRA pre-approves a standard tariff for a technology (e.g. solar, wind) that a generation company can accept on a "take it or leave it" basis, skipping a lengthy case-by-case review.
Competitive Bidding Tariff	Multiple bidders compete for a project, generally through an EOI/RFP process, and the license is awarded to the lowest-cost qualified bidder.

NEPRA first introduced upfront tariffs for solar power generation in 2015, followed by wind power in 2017, under the NEPRA Up-front Tariff (Approval & Procedure) Regulations, 2011 - both of which remain reference points for the upfront-tariff route today. Competitive bidding, by contrast, is governed by the NEPRA Competitive Bidding Tariff (Approval Procedure) Regulations, 2017, which were themselves amended in 2019 (S.R.O. 259(I)/2019) to streamline bid evaluation and benchmark-tariff disclosure.

I also reviewed the regulations governing supporting processes - the NEPRA (Fees Pertaining to Tariff Standards and Procedure) Regulations, 2002, which set the filing fees licensees and consumers pay for tariff petitions and license applications, and the NEPRA (Resolution of Disputes between Independent Power Producers and other Licensees) Regulations, 2003, which allow licensees to refer certain disputes to the Authority. A sample of current generation license fees is shown below:

Generation Licensee Category	Fee (Rs.)
Up to 1 MW	60,000
More than 1 MW up to 10 MW	160,000
More than 10 MW up to 20 MW	240,000

Generation Licensee Category	Fee (Rs.)
More than 50 MW up to 100 MW	480,000
Above 100 MW	800,000
Generation company opting for an upfront tariff	200,000

This closed the loop on the internship's regulatory learning: where Chapters 2 and 4 of the State of the Industry Report showed the sector's performance outcomes, the licensing and tariff framework explains the mechanism NEPRA actually uses to admit new licensees and set the prices that drive those outcomes.

4. Weekly Activity Summary

Week	Dates	Dept.	Key Tasks	Hrs
1	06-10 Jul	IT	Onboarding; NEPRA website familiarization; began Google Search Console (indexing, page submission)	30
2	13-17 Jul	IT	Monitored indexing coverage in Search Console; identified & debugged website bugs; worked on NEPRA Document Intelligence Portal (upload, crawler, search)	30
3	20-24 Jul	Technical	Introduction to Technical dept.; studied AMI meter components; began transformer study	30
4	27-31 Jul	Technical	AMI communication protocols (PLC/cellular); transformer types, ratings & losses	30
5	03-07 Aug	Technical	Transformer protection & testing; reviewed Ch. 2 & Ch. 4 (NTDC) data	30
6	10-14 Aug	Technical	Studied NEPRA licensing & tariff framework; compiled final internship report	30

A detailed, form-based version of this log (following NUST's Anx-C Internship Activity Log format) has been submitted separately for supervisor sign-off.

5. Skills & Learning Outcomes

5.1 Technical Skills

- Practical understanding of Advanced Metering Infrastructure: components, communication protocols, and rollout challenges in the Pakistani context.
- Working knowledge of power transformer construction, ratings, losses, and protection/testing basics.
- Ability to read and analyze a formal regulatory publication (State of the Industry Report) and extract sector-level data and trends.
- Basic understanding of Pakistan's transmission network (NGC/NTDC) - grid station capacity, T&T losses, and investment planning.
- Working knowledge of NEPRA's licensing categories and its three generation tariff-setting mechanisms (cost-plus, upfront, competitive bidding).

5.2 Web & Digital Skills

- Hands-on use of Google Search Console for indexing management and search visibility.

- Practical exposure to website content maintenance and functional debugging on a live, public-facing government website.
- Contributed to a document intelligence portal - file upload, web crawling, and full-text search/indexing of regulatory filings.

5.3 Professional Skills

- Exposure to how a federal regulatory authority is structured and how its departments (IT and Technical) support its regulatory mandate.
- Report writing and technical documentation.

6. Conclusion

This six-week internship at NEPRA provided a balanced mix of digital/IT exposure and core electrical engineering content. The IT Department phase built practical skills in website indexing and debugging, while the Technical Department phase connected classroom knowledge of AMI systems and transformers to their real-world regulatory context, as documented in NEPRA's State of the Industry Report 2025. Overall, the internship strengthened both technical understanding and the ability to work with real regulatory data and documentation.

Site Supervisor (Technical): Attar Rehman, Deputy Director (Technical) Signature: _____

Faculty Supervisor / Dept. ILO Signature: _____

Student-Intern Signature: _____